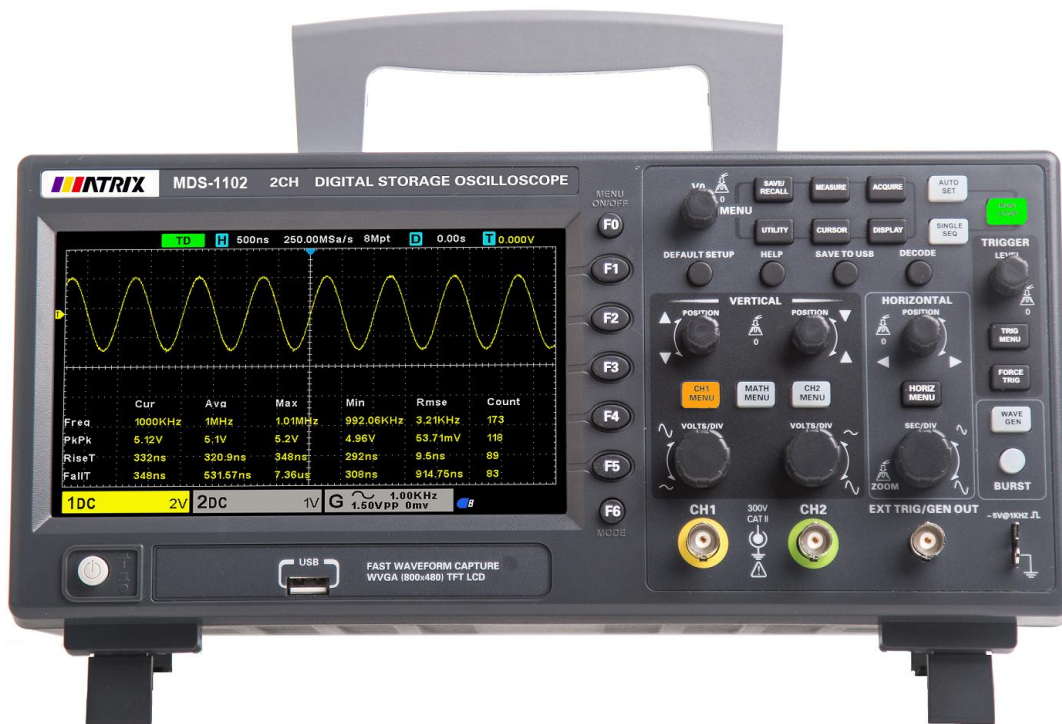


User manual link: <https://wendy4197.github.io/MATRIX/MDS-1000Series-UserManual.pdf>



◆ Features:

- 2 channels which are respectively controlled by independent knobs
- 100 MHz analog channel bandwidth
- Sampling rate up to 1 GSa/s
- 8M memory depth
- Vertical range 2mV/div ~ 10V/div
- Vertical resolution: 8bit
- Trigger: Edge, Pulse, Video, Slope, Overtime, Window, Pattern, Interval, Under Amp, UART, LIN, CAN, SPI, IIC
- BUS decode and protocol analysis: RS232/UART, I2C, SPI, CAN, LIN
- Can save multiple data formats, such as settings, waveforms, reference waveforms, CSV, pictures
- A 3-digit digital voltage meter and a 6-digit hardware frequency indicator
- 32 kinds of auto measurements with statistics, real-time statistics of maximum, minimum, standard deviation and etc.
- 2 sets of digital voltmeters
- Support threshold testing, free measurements within the screen
- Abundant SCPI remote command control
- USB Host/Device.

Model	MDS-1102
Bandwidth	100MHz
Oscilloscope channels	2CH
Oscilloscope	
Sample rate	1GSa/s (single channel) 500MSa/s (two channels)
Acquisition	
Normal	Sample data
Peak-to-peak value	Display high frequency and random burr
Average	Average waveform, times: 4, 8, 16, 32, 64, 128
High resolution	Up to 12bit
Input	
Input coupling	DC, AC, GND
Input impedance	1MΩ±2% 13pF±3pF
Probe attenuation factor	1X, 10X, 100X, 1000X
Voltage rating	300V CAT II
Maximum input voltage	300VRMS (10X)
Horizontal	
Waveform interpolation	(sin x)/x
Maximum record length	Single channel maximum 8M
	Two channels maximum 4M
Horizontal scale range	2ns/div~100s/div 1, 2, 5 step by step
Time base mode	Y-T, X-Y, Roll
Zero offset	±0.5 div×minimum time base gear
Sample Rate and Delay Time Accuracy	±25ppm
Delta Time Measurement Accuracy (Full Bandwidth) Sample Rate and Delay Time Accuracy	single-shot, Normal mode ± (1 sample interval+100ppm×reading+0.6ns)
	> 16 times averages ± (1 sample interval+100ppm×reading+0.4ns)
	Sample interval=sec/div÷200

Sample Rate and Delay Time Accuracy	±50ppm（at any interval greater than 1ms）	
Vertical		
Model	MDS-1102	
Bandwidth	100MHz	
Rising time in BNC position (typical)	3.5ns	
Vertical resolution	8 bits resolution, each channel samples simultaneously	
Vertical sensitivity	2mV/div to 10V/div	
Offset range	≥ 200mV/div, ±1V;	
	<200mV/div ±50V	
Mathematical operation	+, -, ×, ÷, FFT	
FFT	Window: Rectangle, Hanning, Hamming, Blackman, Bartlett, Flattop	
Bandwidth Limit	20MHz	
Bass response (-3db)	In BNC position ≤ 10Hz	
Vertical gain accuracy	In "normal" or "average" acquisition mode, the accuracy of 10V/div to 10mV/div is ±3%;	
	In "normal" or "average" acquisition mode, the accuracy of 5mV/div to 2mV/div is ±4%	
Note: Bandwidth reduced to 6MHz when using a 1X probe		
Trigger		
Trigger type	Edge, Pulse width, Video, Slope, Overtime, Window, Pattern, Interval, Under Amp, UART, LIN, CAN, SPI, IIC	
Trigger level range	±5 divisions from the center of the screen	
Trigger mode	Auto, Normal, single	
Level	CH1~CH2	±4 divisions from the center of the screen
	EXT(Only With AWG Model)	0~3.3V
Holdoff range	8ns~10s	
Trigger level accuracy	CH1~CH2	0.2 div×volts/div within ±4 divisions from the center of the screen

	EXT(Only With AWG Model)	$\pm$ (Set value $\times$ 6%+40mV)
Edge trigger	Slope	Rising edge, falling edge, rising or falling edge
	Signal source	CH1, CH2, EXT(Only With AWG Model)
Pulse width trigger	Polarity	Positive polarity, negative polarity
	Condition (When)	<, >, !=, =
	Signal source	CH1~CH2,
	Pulse width range	8ns ~ 10s
	Accuracy	8ns
Video trigger	Signal standard	NTSC, PAL
	Signal source	CH1~CH2
	Synchronization	Scanning line, line number, odd field, even field, all field
Slope trigger	Slope	rising, falling
	Condition (When)	<, >, !=, =
	Signal source	CH1 ~ CH2
	Time range	8ns ~ 10s
	Accuracy	8ns
Overtime trigger	Signal source	CH1~CH2,
	Polarity	Positive polarity, negative polarity
	Time range	8ns ~ 10s
	Accuracy	8ns
Window trigger	Signal source	CH1~CH2
Pattern trigger	Pattern	0: low level; 1: high level; X: ignore
	Level (signal source)	CH1~CH2

Interval trigger	Slope	rising, falling
	Condition (When)	<, >, !=, =
	Signal source	CH1~CH2
	Time range	8ns ~ 10s
	Accuracy	8ns
Under Amp trigger	Polarity	Positive polarity, negative polarity
	Condition (When)	<, >, !=, =
	Signal source	CH1~CH2
	Time range	8ns ~ 10s
	Accuracy	8ns
UART trigger	Condition (When)	Start, Stop, data, Parity ERR, COM ERR
	Signal source(RX/TX)	CH1~CH2
	Data format	Hex (hexadecimal)
	Data length	1 byte
	Data bit width	5 bit, 6 bit, 7 bit, 8 bit
	Odd-even check	none, odd, even
	Idle level	high, low
	Baud rate (optional)	110/300/600/1200/2400/4800/9600/14400/19200/38400/57600/115200/230400/380400/460400 bit/s
	Baud rate(user-defined)	300bit/s~334000bit/s
LIN trigger	Condition (When)	Interval field, synchronization field, ID field, synchronization error, identifier, ID and data
	Signal source	CH1~CH2
	Data format	Hex (hexadecimal)

	Baud rate (optional)	110/300/600/1200/2400/4800/9600/14400/19200/38400/57600/115200/230400/380400/460400 bit/s
	Baud rate(user-defined)	300bit/s~334000bit/s
CAN trigger	Condition (When)	Start bit, remote frame ID, data frame ID, frame ID, data frame data, error frame, all errors, ACK Error, overload frame
	Signal source	CH1~CH2
	Data format	Hex (hexadecimal)
	Baud rate (optional)	10000, 20000, 33300, 500000, 62500, 83300, 100000, 125000, 250000, 500000, 800000, 1000000
	Baud rate(user-defined)	5kbit/s~1Mbit/s
SPI trigger	Signal source	CH1~CH2
	Data format	Hex (hexadecimal)
	Data bit width	4, 8, 16, 24, 32
IIC trigger	Signal source (SDA/SC L)	CH1~CH2
	Data format	Hex (hexadecimal)
	Data index	0~7
	When(condition)	Start bit, stop bit, No Ack, address, restart, address and data
Measurement		
Cursor	Voltage difference between cursors ΔV	
	Time difference between cursors ΔT	
	Reciprocal of ΔT , in Hertz ($1/\Delta T$)	
Auto measurement	frequency, period, mean, peak-to-peak, RMS, minimum, mixmum, rising time, falling time, + width, - width, base, top, middle, amplitude, overshoot, preshoot, rising edge phase difference, falling edge phase difference, + duty, - duty, period mean, PRMS, FOVshoot, ROVshoot, BWIDTH, FRF, FFR, LRR, LRF, LFR, LFF	
DVM	Data	CH1, CH2

	source	
	Measurement type	DC RMS
		AC RMS
		DC
	Frequency meter	hardware 6 bits frequency meter
General specifications		
Display	Display type	7" diagonal TFT liquid crystal
	Display resolution	800 (horizontal)*480 (vertical) pixels
	Display colour	16 million colours (24 bits true colour)
	Persistence time	minimum, 1 s, 5 s, 10 s, 30 s, infinite
	Display type	dot, vector
	Display brightness	adjustable
	Grid type	adjustable
	Grid brightness	adjustable
Interface	Standard interface	USB Host, USB Device
General specifications	Probe compensator output	
	Output voltage, typical	about 2Vpp input $\geq 1\text{M}\Omega$ load
	Frequency, typical	1kHz
	Power supply	100-120VAC _{RMS} ($\pm 10\%$), 45Hz to 440Hz, CAT II
		120-240VAC _{RMS} ($\pm 10\%$), 45Hz to 66Hz, CAT II
	Power consumption	<30W
	Fuse	T, 3.15A, 250V, 5x20mm
	Operating temperature	0~50 °C (32~122 °F)

	Storage temperature	-40~+71 °C (-40~159.8 °F)	
	Humidity	≤+104°F (≤+40°C): ≤90% relative humidity	
		106°F ~122°F (+41°C ~50°C): ≤60% relative humidity	
	Altitude	Operating and nonoperating	3, 000m (10, 000 feet)
	Mechanical shock	Random vibration	0.31 g _{RMS} from 50Hz to 500Hz,
			10 minutes on each axis
		Nonoperating	2.46g _{RMS} from 5Hz to 500Hz,
			10 minutes on each axis
		Operating	50g, 11ms, half-sine wave
	Mechanical	Size	318 x 110 x 150mm (length x width x height)
		Weight	1900g